

Level Zero support, extending its compatibility from ageing Broadwell to state-of-the-art DG2 discrete graphics and the latest Meteor Lake processors, featuring improved integrated graphics.

The 23.39.27427.23 isn't just a routine update; it brings crucial fixes and low-level code refinements, ensuring a seamless experience across diverse Linux distributions.

Although a concise change-log is absent, a closer look reveals over 230 new code commits, encompassing performance optimisations and fixes for Linux 6.8's Xe kernel mode driver. The update introduces Ray Tracing Acceleration Structure (RTAS) support, Meteor Lake updates, and routine code refinements.

For eager users, the open source code and pre-built Debian/Ubuntu binaries are available on GitHub, underscoring Intel's commitment to an inclusive development environment.

In tandem with the release, Intel ensures compatibility with recent graphics generations, supporting OpenCL 3.0 and oneAPI Level Zero 1.3.

MOCAT to address space debris concerns

Against the backdrop of mounting concerns over space debris and collisions, MIT's Astrodynamics, Space Robotics, and Controls Laboratory (ARCLab) has introduced the MIT Orbital Capacity Assessment Tool (MOCAT). Unveiled during the 2023 Organisation for Economic Cooperation and Development (OECD) Space Forum



Workshop in December last year, the public beta release aims to address the surging challenges in low Earth orbit.

Amidst the increasing congestion driven by a satellite deployment surge, MOCAT becomes a crucial open source model. It allows users to simulate the long-term space environment, offering insights into space debris growth and the effectiveness of preventive measures.

Richard Linares, principal investigator

for MOCAT and an MIT associate professor of aeronautics and astronautics, remarked on its significance: "MOCAT represents a significant leap forward in orbital capacity assessment. By making it open source and publicly available, we hope to engage the global community in advancing our understanding of satellite orbits and contributing to the sustainable use of space."

The tool, developed over several years, is intended to empower stakeholders, including satellite operators and regulators, to make data-driven decisions. This strategic beta release encourages user experimentation and feedback to guide further development.

Supported by DARPA and NASA's Office of Technology and Strategy, MOCAT's open source approach was hailed by Charity Weeden, associate administrator for the Office of Technology, Policy, and Strategy at NASA headquarters: "This open source modelling tool is a public good that will advance space sustainability, improve evidence-based policy analysis, and help all users of space make better decisions."

Libreboot now supports HP EliteBook 820 G2 laptops

The Libreboot project has released a new patch, expanding its roster of supported mainboards to include the HP EliteBook 820 G2.

The HP EliteBook 820 G2 is a Broadwell laptop, offering various configurations powered by Intel processors such as the Core i5-5200U, Core i5-5300U, Core i7-5500U, or Core i7-5600U. Notable features include three USB 3.0 ports, a 4K 60Hz-capable DisplayPort, a VGA port, support for up to 32GB of RAM, and three storage slots accommodating PCIe, M.2, and regular SATA drives.

Libreboot, known for its commitment to providing open source alternatives for BIOS/UEFI firmware, recently updated its release to include support for various mainboards, such as the HP EliteBook 2570p, HP 8300 USDT, and Gigabyte GA-G41M-ES2L.

Those interested in supporting the project can acquire an HP EliteBook 820 G2 with Libreboot already integrated. This opportunity serves a dual purpose, as the company offering these laptops is owned by one of the developers, contributing directly to Libreboot's ongoing initiatives.

For enthusiasts who prefer a hands-on approach, the option to install Libreboot independently on an existing HP EliteBook 820 G2 is also available. A comprehensive hardware/install guide is provided to assist users through the installation process.

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